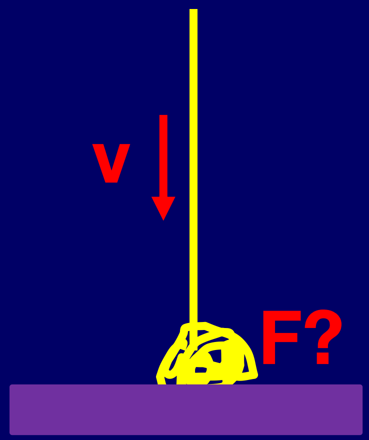
Quiz 5: If you first hold the upper end of a uniform soft chain while the lower end just touches the floor. After you release, how large force will the chain impose on the floor? Assuming the chain has the length L and the mass M.

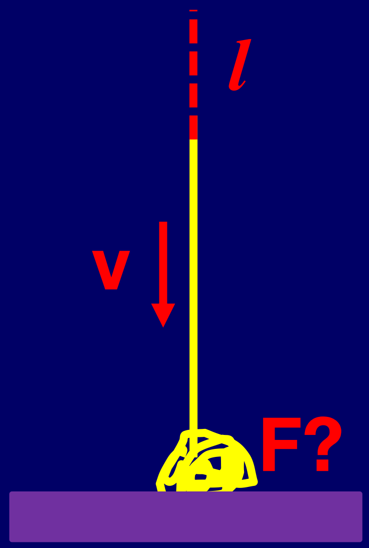


Answer: At the moment t, the part of the chain with the length l has laid on the floor. During the following interval dt, another small part with mass dm will hit the floor.

Then, the change of the momentum of this small part will be

$$dp = dm \cdot v = (M/L)dl \cdot v = (M/L)vdt \cdot v$$

where $v = \sqrt{2gl}$



From $F_p dt = dp$, we have $F_p = 2gMl/L$

Thus the total force at the time t is

$$F_t = F_l + F_p = Mgl/L + 2gMl/L$$

$$= 3gMl/L = 3mg$$

where $m = Ml/L$